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Communication device

Abstract

A communication device includes a body, and a wireless communicator housed in the body for communicating wirelessly with an external device proximate to the body through magnetic coupling. The body is placed in an opening in a wall surface and fixed to the wall surface with an exposed surface of the body exposed from the wall surface. The wireless communicator includes a substrate including a coil pattern to allow the magnetic coupling with the external device. The substrate is nonparallel to the exposed surface.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION

(1) This application is a national stage application, pursuant to 35 U.S.C. § 371, of International Patent Application No. PCT/JP2022/044445, filed Dec. 1, 2022, the entire contents of which are incorporated herein by reference.

TECHNICAL FIELD

(2) The present disclosure relates to a communication device.

BACKGROUND ART

(3) In known equipment such as a control panel used at a factory automation (FA) site, a device may be installed to provide additional functions such as a user interface. When the device installed in the equipment can wirelessly communicate with a mobile terminal such as a smartphone or a tablet terminal, the equipment is more convenient, allowing easy data input or output by the operator.

(4) For equipment users to easily install and use the device without complicated preliminary work such as network setting, the device installable in the equipment may perform short-range wireless communication with a mobile terminal. To achieve this, short-range wireless communication techniques may be applied to the device installable in the equipment (see, for example, Patent Literature 1).

CITATION LIST

Patent Literature

(5) Patent Literature 1: Unexamined Japanese Patent Application Publication No. 2011-091503

SUMMARY OF INVENTION

Technical Problem

(6) Patent Literature 1 describes a digital camera including an antenna that performs short-range wireless communication when an antenna in another device is proximate to the camera. Such a digital camera can be approached by another device from above, below, right, left, front, and rear and thus have a high degree of flexibility in the installation of the antenna.

(7) However, the device installed in the equipment is typically embedded in and integral with the equipment except a surface for providing the functions of the device, thus performing short-range wireless communication through the surface. When a flat antenna is incorporated to face the antenna of another device as described in Patent Literature 1, the surface of the device installable in the equipment is to have a large area corresponding to at least the area of the antenna. This may increase the size of the device when the function of short-range wireless communication is implemented in the device installable in the equipment.

(8) Under such circumstances, an objective of the present disclosure is to allow a device installable in equipment to implement short-range wireless communication without increasing the size of the device.

Solution to Problem

(9) To achieve the above objective, a communication device according to an aspect of the present disclosure includes a body, and wireless communication means housed in the body for communicating wirelessly with an external device proximate to the body through magnetic coupling. The body is placed in an opening in a wall surface and fixed to the wall surface with an exposed surface of the body exposed from the wall surface. The wireless communication means includes a substrate including a coil pattern to allow the magnetic coupling with the external device. The substrate is nonparallel to the exposed surface.

Advantageous Effects of Invention

(10) In the structure according to the above aspect of the present disclosure, the exposed surface is nonparallel to the substrate including the coil pattern. The region to be used for wireless communication in the exposed surface is thus smaller than the area of the substrate. This allows a device installable in equipment to implement short-range wireless communication without increasing the size of the device.

Description

BRIEF DESCRIPTION OF DRAWINGS

(1) FIG. **1** is an external view of a communication device according to Embodiment 1;

(2) FIG. **2** is a diagram of the communication device according to Embodiment 1 installed in equipment;
(3) FIG. **3** is a block diagram of a body in Embodiment 1;
(4) FIG. **4** is a schematic diagram of a magnetic field generated by a coil pattern in Embodiment 1;
(5) FIG. **5** is a diagram of a substrate in Embodiment 1 inclined with respect to an exposed surface;
(6) FIG. **6** is a diagram illustrating a frame including a substrate in a comparative example;
(7) FIG. **7** is a diagram illustrating a coil pattern in Embodiment 2;
(8) FIG. **8** is a schematic diagram of a magnetic field generated by the coil pattern in Embodiment 2;
(9) FIG. **9** is a first diagram illustrating a coil pattern in a modification;
(10) FIG. **10** is a diagram illustrating a coil pattern in Embodiment 3;
(11) FIG. **11** is a schematic cross-sectional view of a substrate in Embodiment 3;
(12) FIG. **12** is a schematic diagram of a magnetic field generated by the coil pattern in Embodiment 3;
(13) FIG. **13** is a schematic diagram of a generated magnetic field in a modification;
(14) FIG. **14** is a schematic first cross-sectional view of a substrate in a modification;
(15) FIG. **15** is a second schematic cross-sectional view of a substrate in the modification;
(16) FIG. **16** is a diagram illustrating a coil pattern in Embodiment 4;
(17) FIG. **17** is a schematic cross-sectional view of a substrate in Embodiment 4;
(18) FIG. **18** is a schematic third cross-sectional view of a substrate in the modification;
(19) FIG. **19** is a schematic fourth cross-sectional view of a substrate in the modification;
(20) FIG. **20** is a diagram illustrating a coil pattern in Embodiment 5;
(21) FIG. **21** is a schematic cross-sectional view of a substrate in Embodiment 5;
(22) FIG. **22** is a schematic fifth cross-sectional view of a substrate in the modification; and
(23) FIG. **23** is a diagram of a communication device according to a modification.

DESCRIPTION OF EMBODIMENTS

(24) A communication device according to one or more embodiments of the present disclosure is described below in detail with reference to the drawings. An arrow Z**1** indicates a direction in which the communication device is placed in equipment, and an arrow Z**2** indicates the direction opposite to the arrow Z**1** in the drawings. Arrows X**1**, X**2**, Y**1**, and Y**2** each indicate a direction perpendicular to the arrows Z**1** and Z**2**. The one or more embodiments are described focusing on examples in which the arrows X**1** and X**2** point horizontally and the arrows Y**1** and Y**2** point vertically. The exposed surface of the communication device installed in the equipment corresponds to a plane X-Y.

Embodiment 1

(25) As illustrated in FIG. **1**, a communication device **100** according to the present embodiment is a display installed in equipment **200** that is a control panel. The communication device **100** functions as a user interface for the equipment **200**. The communication device **100** includes a body **101** that provides the user interface function of the communication device **100**, attachments **102** that fix the body **101** to the equipment **200**, and an exposed surface **103** fixed to the equipment **200** to face outside the equipment **200**.

(26) The communication device **100** is installed in the equipment **200** with the body **101** placed along the arrow Z**1** into an opening **201** in a wall surface **203** of the equipment **200**. The surface of the communication device **100** opposite to the exposed surface **103** is placed first into the opening **201**. The attachments **102** are then fastened to holes **202** near inner wall surfaces of the equipment **200** with screws (not illustrated). Although FIG. **1** illustrates the attachments **102** as metal fittings each with a threaded hole, the attachments **102** may be metal fitting holders to which such metal fittings are attachable after the body **101** is placed in the opening **201**. The body **101** is placed in the opening **201** in the wall surface **203** and fixed to the wall surface **203** with the exposed surface **103** being a surface of the body **101** in the Z**2** direction exposed from the wall surface **203**. The

exposed surface **103** includes a display screen **104** that displays an image for an operator who operates the equipment **200**.
(27) In the communication device **100**, a substrate **132** including a coil pattern that functions as an antenna for short-range wireless communication is placed to be parallel to a plane X-Z. FIG. **2** illustrates the communication device **100** installed in the equipment **200** and integral with the equipment **200**. As illustrated in FIG. **2**, the exposed surface **103** includes the display screen **104** and a frame for the display screen **104**.
(28) The communication device **100** is a display that functions mainly as a user interface. The display screen **104** that provides the function may thus use a large proportion of the exposed surface **103** with the frame being small. The operator places a mobile terminal proximate to the substrate **132** located inside the frame for data input or output through short-range wireless communication.
(29) FIG. **3** is a block diagram of the body **101**. As illustrated in FIG. **3**, the body **101** includes a controller **110** that controls the components of the body **101**, a device communicator **120** that communicates with the equipment **200**, a wireless communicator **130** that performs short-range wireless communication with an external device, and a display **140** that displays an image on the display screen **104**. The display screen **104** corresponds to an example of a screen on which an image appears, and the display **140** corresponds to an example of display means for displaying an image on the screen in the exposed surface.
(30) The device communicator **120** may be connected to the equipment **200** with a cable for wired communication with the equipment **200** or perform wireless communication with the equipment **200**.
(31) The controller **110** includes a central processing unit (CPU) or a micro processing unit (MPU) and memories such as a random-access memory (RAM) and a read-only memory (ROM). The controller **110** executes a program stored in such memories to control the device communicator **120**, the wireless communicator **130**, and the display **140**.
(32) More specifically, the controller **110** functions as an input interface for the communication device **100** by providing the operator's operation on a touchscreen of the display **140** or the results of processing the operation to the equipment **200** through the device communicator **120**. The controller **110** also functions as an output interface for the communication device **100** by producing an image based on a signal received from the equipment **200** through the device communicator **120** and displaying the image on the display **140**. The controller **110** further processes data received from an external device through the wireless communicator **130** or transmits data to the external device through the wireless communicator **130**.
(33) The wireless communicator **130** corresponds to an example of wireless communication means housed in the body **101** for communicating wirelessly with an external device proximate to the exposed surface **103** of the body **101** through magnetic coupling. The wireless communicator **130** includes a communication circuit **131** including a communication interface circuit and the substrate **132** including a coil pattern **133** to allow the magnetic coupling with the external device. The coil pattern **133** includes a conductor indicated by a solid line in FIG. **3** and printed on the front surface of the substrate **132** that is a rigid substrate, and a conductor indicated by a dashed line and printed on the rear surface of the substrate **132**. These conductors are electrically connected together with a through-hole or with a conductor extending through an insulator layer in the substrate. The coil pattern **133** thus forms a coil and functions as an inductor. The communication circuit **131** transmits and receives data to and from the external device through magnetic coupling between the coil pattern **133** and a coil included in the external device.
(34) With the substrate **132** placed to be parallel to the plane X-Z as described above, a current flowing in the coil pattern **133** generates a magnetic field as indicated by dashed lines in FIG. **4**. Although the magnetic flux density produced by a coil usually has a large value at the central axis of the coil, a fringe magnetic field extends to an external device **300** located in the Z**2** direction as

illustrated in FIG. **4**, thus allowing communication with the coil pattern **133** used as an antenna.
(35) The magnetic field produced from an antenna in the external device **300** also induces a current although the current is weaker than a current generated when substrates including antennas face each other, thus allowing the wireless communicator **130** to receive data from the external device **300**. When the central axis of the coil serving as the antenna in the external device **300** is aligned with the substrate **132**, almost no current is induced. In this case, a message may be displayed on the display **140** to prompt the operator to adjust the position or orientation of the external device **300**.
(36) In this example, the substrate **132** described above is parallel to the plane X-Z, or more specifically, the substrate **132** is perpendicular to the exposed surface **103**, with the substrate **132** forming an angle θ of 90 degrees with the exposed surface **103**. However, the present disclosure is not limited to the example. As illustrated in FIG. **5**, the angle θ may be greater than 0 degrees and less than 90 degrees. In other words, the substrate **132** and the exposed surface **103** are to be nonparallel. As illustrated in FIG. **5**, the angle θ between the substrate **132** and the exposed surface **103** may be the angle between the extended plane of the substrate **132** and the extended plane of the exposed surface **103**. As illustrated in FIGS. **4** and **5**, the angle θ may be equal to an acute angle between the substrate **132** and the exposed surface **103**.
(37) As described above, the exposed surface **103** is nonparallel to the substrate **132** including the coil pattern **133** as the antenna. In the exposed surface **103**, the region to be used for wireless communication is thus smaller than the area of the substrate. For example, a width D**1** of the frame surrounding the display screen in the exposed surface **103** illustrated in FIG. **5** is smaller than a width D**2** of a frame on an exposed surface **103** parallel to a substrate **132***a* illustrated as a comparative example in FIG. **6**. This allows the function of short-range wireless communication to be implemented in the communication device **100** installable in the equipment **200** without increasing the size of the communication device **100**.
(38) In particular, when the angle θ is 60 degrees or more, the region to be used in the exposed surface **103** may be equal to or less than half of the area of the substrate **132**. More specifically, when the angle θ is 85 degrees or more, the region to be used in the exposed surface **103** may be equal to or less than 10% of the area of the substrate **132**.
(39) Although the coil pattern **133** described above includes two windings, the coil pattern **133** may include a single winding or two or more windings.
Embodiment 2
(40) Embodiment 2 is described focusing on the differences from Embodiment 1 described above. Like reference signs denote like or corresponding components in Embodiment 1. The present embodiment differs from Embodiment 1 in that the coil pattern **133** includes multiple coil patterns with different center points.
(41) As illustrated in FIG. **7**, the coil pattern **133** includes a first coil pattern **151** indicated by a solid area and a second coil pattern **152** indicated by a hatched area. The first coil pattern **151** and the second coil pattern **152** each include a single winding, and the coil pattern **133** corresponds to a coil with two windings. The first coil pattern **151** and the second coil pattern **152** are located on the same surface **1321** of the substrate **132**, with the first coil pattern **151** located outward from the second coil pattern **152**.
(42) The second coil pattern **152** has a center point C**22** closer to the exposed surface **103** than a center point C**21** of the first coil pattern **151**. The center points C**21** and C**22** may be the gravity centers of the first coil pattern **151** and the second coil pattern **152** or the points at which the magnetic flux density is greatest when a current flows in each of the first coil pattern **151** and the second coil pattern **152** on the surface **1321** of the substrate **132** including the first coil pattern **151** and the second coil pattern **152**.
(43) The first coil pattern **151** includes a first conductor **151***a* adjacent to the exposed surface **103** and a second conductor **151***b* opposite to the exposed surface **103** across the first conductor **151***a*.

The second coil pattern **152** includes a third conductor **152***a* adjacent to the exposed surface **103** and a fourth conductor **152***b* opposite to the exposed surface **103** across the third conductor **152***a*. A distance D**11** between the first conductor **151***a* and the third conductor **152***a* is smaller than a distance D**12** between the second conductor **151***b* and the fourth conductor **152***b*.
(44) In the conductor patterns in the coil pattern **133** in the present embodiment, the conductor patterns adjacent to the exposed surface **103** are closer to each other as illustrated in FIG. **8**. This may increase the surrounding magnetic flux density, thus improving the efficiency of short-range wireless communication with the external device **300**.
(45) Although the first coil pattern **151** and the second coil pattern **152** described above each include a single winding, at least one of the first coil pattern **151** or the second coil pattern **152** may include two or more windings. FIG. **9** illustrates a coil pattern **133** including a first coil pattern **151** and a second coil pattern **152** each including two windings. For the first coil pattern **151** and the second coil pattern **152** with any number of windings, a distance D**22** between the exposed surface **103** and the center point C**22** of the second coil pattern **152** is to be smaller than a distance D**21** between the exposed surface **103** and the center point C**21** of the first coil pattern **151**. For the first coil pattern **151** located inward with the second coil pattern **152** located outward, the distance D**21** is to be smaller than the distance D**22**. The coil pattern **133** may include the first coil pattern **151** and the second coil pattern **152** with the distance D**21** different from the distance D**22**. The distance D**21** corresponds to an example of a first distance between an exposed surface and the center of a first coil pattern. The distance D**22** corresponds to an example of a second distance between the exposed surface and the center of a second coil pattern.
Embodiment 3
(46) Embodiment 3 is described focusing on the differences from Embodiment 1. Like reference signs denote like or corresponding components in Embodiment 1. The present embodiment differs from Embodiment 1 in that the coil pattern **133** includes multiple coil patterns located on different layers in the substrate **132**.
(47) As illustrated in FIG. **10**, the coil pattern **133** includes a first coil pattern **161** indicated by a solid area and a second coil pattern **162** indicated by a hatched area. The first coil pattern **161** is located on the front surface of the substrate **132** in the Y**1** direction. The second coil pattern **162** is located on the rear surface of the substrate **132** in the Y**2** direction and is electrically connected to the first coil pattern **161** with an electrical conductor **163***a* that is a through-hole conductor or a via conductor. The first coil pattern **161** and the second coil pattern **162** each include a single winding, and the coil pattern **133** corresponds to a coil with two windings.
(48) The first coil pattern **161** has a center point C**31** closer to the exposed surface **103** than a center point C**32** of the second coil pattern **162**. More specifically, a distance D**31** between the exposed surface **103** and the center point C**31** is smaller than a distance D**32** between the exposed surface **103** and the center point C**32**. The center points C**31** and C**32** may be defined in the same manner as the center points C**21** and C**22** in Embodiment 2.
(49) FIG. **11** is a schematic cross-sectional view of the substrate **132** taken along line A-A. As illustrated in FIG. **11**, the first coil pattern **161** is located on the conductor layer at the front surface of the substrate **132**, and the second coil pattern **162** is located on the conductor layer at the rear surface of the substrate **132**. As illustrated in FIG. **11**, the coil corresponding to the coil pattern **133** including the first coil pattern **161** and the second coil pattern **162** is inclined with respect to the substrate **132**.
(50) In the communication device **100** including the coil pattern **133** according to the present embodiment, the central axis of the coil pattern **133** is inclined with respect to a Y-axis although the substrate **132** and the exposed surface **103** form an angle θ of 90 degrees as illustrated in FIG. **12**. The magnetic flux density value is thus expected to be greater at the position of the external device **300**. This may improve the efficiency of communication with the external device **300**.
(51) When the angle θ is less than 90 degrees as illustrated in FIG. **13**, the central axis of the coil

comes more proximate to a Z-axis, and the efficiency of communication with the external device **300** is expected to be improved further.

(52) The substrate **132** may be a multilayered substrate with the coil pattern **133** including coil patterns located on different conductor layers. FIG. **14** illustrates an example in which the coil pattern **133** includes the first coil pattern **161** and the second coil pattern **162** as well as a third coil pattern **163** located in the Y**1** direction from the first coil pattern **161**, and a fourth coil pattern **164** and a fifth coil pattern **165** located in the Y**2** direction from the second coil pattern **162**. Each coil pattern has the same distance D**30** between the conductor adjacent to the exposed surface **103** and the opposite conductor. The center points C**31** and C**32**, a center point C**33** of the third coil pattern **163**, a center point C**34** of the fourth coil pattern **164**, and a center point C**35** of the fifth coil pattern **165** are closer to the exposed surface **103** in the order along the arrow Y**1**. In other words, the center points C**31**, C**32**, C**33**, C**34**, and C**35** in the individual layers are closer to the exposed surface **103** in the stacking direction of the layers in the substrate **132**.

(53) The arrangement in FIG. **14** may further be modified as illustrated in FIG. **15**. In the example in FIG. **15**, the center point C**33** is closer to the exposed surface **103** than the center point C**31**, the center point C**34** is closer to the exposed surface **103** than the center point C**32**, and the center point C**35** is closer to the exposed surface **103** than the center point C**34**.

(54) Although the first coil pattern **161** and the second coil pattern **162** described above each include a single winding, at least one of the first coil pattern **161** or the second coil pattern **162** may include two or more windings.

Embodiment 4

(55) Embodiment 4 is described focusing on the differences from Embodiment 1. Like reference signs denote like or corresponding components in Embodiment 1. The present embodiment differs from Embodiment 1 in that the coil patterns in the coil pattern **133** have conductors adjacent to the exposed surface **103** on the same surface of the substrate **132** and the opposite conductors on different layers.

(56) As illustrated in FIG. **16**, the coil pattern **133** includes a first coil pattern **171** indicated by a solid area and a second coil pattern **172** indicated by a hatched area. The first coil pattern **171** and the second coil pattern **172** each include a single winding, and the coil pattern **133** corresponds to a coil with two windings.

(57) The first coil pattern **171** includes a first conductor **171***a* adjacent to the exposed surface **103** and a second conductor **171***b* opposite to the exposed surface **103** across the first conductor **171***a*. The first conductor **171***a* and the second conductor **171***b* are both located on the surface **1321** that is the front surface of the substrate **132**. The second coil pattern **172** includes a third conductor **172***a* adjacent to the exposed surface **103** and a fourth conductor **172***b* opposite to the exposed surface **103** across the third conductor **172***a*. The third conductor **172***a* is located on the surface **1321** in the same manner as the first coil pattern **171**, whereas the fourth conductor **172***b* is located on the rear surface of the substrate **132**. The third conductor **172***a* and the fourth conductor **172***b* are electrically connected to each other with conductors indicated by outlined circles in FIG. **16**.

(58) FIG. **17** is a schematic cross-sectional view of the substrate **132** taken along line B-B. As illustrated in FIG. **17**, the first conductor **171***a* and the second conductor **171***b* in the first coil pattern **171** are both located on the conductor layer at the front surface of the substrate **132**. The third conductor **172***a* in the second coil pattern **172** is located on the conductor layer at the front surface of the substrate **132**, whereas the fourth conductor **172***b* is located on the conductor layer at the rear surface of the substrate **132**.

(59) The communication device **100** including the substrate **132** according to the present embodiment is expected to have a greater magnetic flux density value at a position adjacent to the exposed surface **103** within the magnetic field produced by the coil pattern **133**, thus improving the efficiency of communication. In particular, the first conductor **171***a* and the third conductor **172***a* may be proximate to each other, and the second conductor **171***b* and the fourth conductor **172***b* may

overlap each other in the plane X-Z.
(60) Although the angle θ described above is 90 degrees, the angle θ may be changed as in the above embodiment.
(61) The substrate **132** may be a multilayered substrate with the coil pattern **133** including more than two coil patterns partially located on different layers. In the example in FIG. **18**, the first conductor **171***a* and the third conductor **172***a* as well as a fifth conductor **173***a*, a seventh conductor **174***a*, and a ninth conductor **175***a* are located adjacent to the exposed surface **103** on the same conductor layer. The second conductor **171***b*, the fourth conductor **172***b*, a sixth conductor **173***b* included in a third coil pattern with the fifth conductor **173***a*, an eighth conductor **174***b* included in a fourth coil pattern with the seventh conductor **174***a*, and a tenth conductor **175***b* included in a fifth coil pattern with the ninth conductor **175***a* are located in the Z**1** direction along the Y-axis. The substrate **132** with this structure may also improve the efficiency of communication. The example in FIG. **18** corresponds to another example of a modification with a typical cylindrical coil bent as illustrated in FIG. **15**.
(62) Although the first coil pattern **171** and the second coil pattern **172** described above each include a single winding, at least one of the first coil pattern **171** or the second coil pattern **172** may include two or more windings.
(63) The conductors located in the Z**1** direction and the conductors located in the Z**2** direction illustrated in FIG. **18** may be replaced with each other. As illustrated in FIG. **19**, the conductors on multiple layers may be located near the exposed surface **103** to produce a magnetic field more uniform at a position adjacent to the exposed surface **103** than at the opposite position, improving the stability of communication with the external device **300**. More specifically, the substrate **132** may have coil patterns symmetric with respect to the X-axis in FIGS. **16** to **18**. The magnetic flux density is affected by various parameters including coil patterns, the distance between the substrate **132** and the exposed surface **103**, materials for the communication device **100** and the equipment **200**, the orientation of the substrate **132**, and other factors. The coil patterns to be included in the communication device **100** may thus be determined after examination of the efficiency and stability of communication.
(64) The coil pattern located on the substrate **132** includes the first coil pattern and the second coil pattern. Each of the first coil pattern and the second coil pattern includes at least one winding. The first coil pattern includes a first conductor located in a first direction and a second conductor located in a second direction. The first direction is one of an outward direction, or the Z**2** direction, in which the exposed surface faces or a direction, or the Z**1** direction, in which the body **101** is placed. The second direction is the other of the directions. The second coil pattern includes a third conductor located in the first direction and a fourth conductor located in the second direction. The first conductor and the third conductor are located on the same surface of the substrate, whereas the second conductor and the fourth conductor are located on different layers in the substrate.
Embodiment 5
(65) Embodiment 5 is described focusing on the differences from Embodiment 1. Like reference signs denote like or corresponding components in Embodiment 1. The present embodiment differs from Embodiment 1 in that some of the conductors in the coil pattern **133** are connected in parallel.
(66) As illustrated in FIG. **20**, the coil pattern **133** includes a first conductor **181** located in the Z**2** direction, or adjacent to the exposed surface **103**, and a conductor **182** located in the Z**1** direction, or opposite to the exposed surface **103** across the first conductor **181**. The conductor **182** includes a second conductor **182***a*, a third conductor **182***b*, and a fourth conductor **182***c* located on different layers. The second conductor **182***a*, the third conductor **182***b*, and the fourth conductor **182***c* are electrically connected parallel to the first conductor **181** with a through-hole conductor or a via conductor.
(67) FIG. **21** is a schematic cross-sectional view of the substrate **132** taken along line C-C. In the example in FIG. **21**, the conductor **182** in the coil pattern **133** electrically connects different

conductor layers with a via conductor.
(68) The communication device **100** including the substrate **132** according to the present embodiment is expected to have a greater magnetic flux density value at a position adjacent to the exposed surface **103** within the magnetic field produced by the coil pattern **133**, thus improving the efficiency of communication.
(69) Although the conductor **182** in the coil pattern **133** described above has conductor patterns located on three different layers, the conductor patterns in the conductor **182** may be located on two or more than three layers. In particular, when the conductor patterns in the conductor **182** is located on five layers, the conductor patterns corresponds to a modified example of a single conductor pattern collectively including the first conductor **171***a*, the third conductor **172***a*, the fifth conductor **173***a*, the seventh conductor **174***a*, and the ninth conductor **175***a* illustrated in FIG. **18**.
(70) Although the coil pattern **133** described above includes a single winding, the coil pattern **133** may include two or more windings.
(71) The conductor located in the Z**1** direction and the conductor located in the Z**2** direction illustrated in FIG. **21** may be replaced with each other. As illustrated in FIG. **22**, the conductors on multiple layers may be located near the exposed surface **103** to produce a magnetic field more uniform at a position adjacent to the exposed surface **103** than at the opposite position, improving the stability of communication with the external device **300**. More specifically, the substrate **132** may have coil patterns symmetric with respect to the X-axis in FIGS. **20** and **21**. The magnetic flux density is affected by various parameters, and the coil patterns to be included in the communication device **100** may thus be determined after examination of the relationship between the parameters and the shape of the coil pattern, and the efficiency and stability of communication.
(72) The coil pattern includes a first conductor located in a first direction and a second conductor and a third conductor located in a second direction and connected parallel to the first conductor. The first direction is one of an outward direction, or the Z**2** direction, in which the exposed surface **103** faces or a direction, or the Z**1** direction, in which the body **101** is placed. The second conductor and the third conductor are located on different layers in the substrate.
(73) Although one or more embodiments of the present disclosure have been described above, the present disclosure is not limited to the above embodiments.
(74) For example, although the communication device **100** described above is substantially the same as the body **101**, the communication device **100** may include components different from the attachments **102** in addition to the body **101**.
(75) The communication device **100** may have any shape other than a rectangular shape. For example, as illustrated in FIG. **23**, the body **101** of the communication device **100** may include a screen unit **105** with a display screen larger than the opening **201** as viewed from the operator located in the Z**2** direction. In this example, when the communication device **100** is placed in the opening **201** in the equipment **200** and installed, the substrate **132** located inside a frame surrounding the display screen of the screen unit **105** achieves short-range wireless communication.
(76) Although the communication device **100** described above is a display, for example, the communication device **100** may be a speaker or a storage installed in the equipment **200** that is a control panel. The equipment **200** may not be a control panel. For example, the equipment **200** may be a wall of a building constructed from a material such as concrete, timber, or metal. The equipment **200** is to have the wall surface **203** to which the communication device **100** is fixed, and the communication device **100** may be fitted into a recess in the wall surface **203**.
(77) Although the communication device **100** described above is being screwed to the equipment **200** with the attachments **102**, the communication device **100** may be installed in the equipment **200** in any other manner. For example, the communication device **100** may not include the attachments **102**. More specifically, attachments may be located on the wall surface **203** or in the opening **201** to fix the rectangular body **101** placed in the opening **201** by pressing or holding the body **101** with the attachments. The communication device **100** may be fixed in the opening **201**

with a magnet located at the bottom of the opening **201** attracting a steel plate included in the housing of the body **101**.

(78) The foregoing describes some example embodiments for explanatory purposes. Although the foregoing discussion has presented specific embodiments, persons skilled in the art will recognize that changes may be made in form and detail without departing from the broader spirit and scope of the invention. Accordingly, the specification and drawings are to be regarded in an illustrative rather than a restrictive sense. This detailed description, therefore, is not to be taken in a limiting sense, and the scope of the invention is defined only by the included claims, along with the full range of equivalents to which such claims are entitled.

INDUSTRIAL APPLICABILITY

(79) One or more embodiments of the present disclosure is applicable to a device fixed to a wall surface for short-range wireless communication.

REFERENCE SIGNS LIST

(80) **100** Communication device **101** Body **102** Attachment **103** Exposed surface **104** Display screen **105** Screen unit **110** Controller **120** Device communicator **130** Wireless communicator **131** Communication circuit **132**, **132***a* Substrate **1321** Surface **133** Coil pattern **140** Display **151**, **161**, **171** First coil pattern **151***a*, **171***a*, **181** First conductor **151***b*, **171***b*, **182***a* Second conductor **152**, **162**, **172** Second coil pattern **152***a*, **172***a*, **182***b* Third conductor **152***b*, **172***b*, **182***c* Fourth conductor **163** Third coil pattern **163***a* Electrical conductor **164** Fourth coil pattern **165** Fifth coil pattern **173***a* Fifth conductor **173***b* Sixth conductor **174***a* Seventh conductor **174***b* Eighth conductor **175***a* Ninth conductor **175***b* Tenth conductor **182** Conductor **200** Equipment **201** Opening **202** Hole **300** External device C**21**, C**22**, C**31**, C**32**, C**33**, C**34**, C**35** Center point D**1**, D**2** Width D**11**, D**12**, D**21**, D**22**, D**30**, D**31**, D**32** Distance θ Angle

Claims

1. A communication device, comprising: a body; and wireless communication circuitry housed in the body and configured to communicate wirelessly with an external device proximate to the body through magnetic coupling, wherein the body is placed in an opening in a wall surface and fixed to the wall surface with an exposed surface of the body exposed from the wall surface, the wireless communication circuitry includes a substrate including a coil to allow the magnetic coupling with the external device, the substrate is nonparallel to the exposed surface, the communication device further comprises a display configured to display an image on a screen thereof within the exposed surface, and the substrate is located inside a frame surrounding the screen.
2. The communication device according to claim 1, wherein the substrate is perpendicular to the exposed surface.
3. The communication device according to claim 1, wherein the coil pattern includes a first coil pattern and a second coil pattern, and the first coil pattern is located on a front surface of the substrate, the second coil pattern is located on a rear surface of the substrate that is opposite the front surface.
4. The communication device according to claim 3, wherein the first coil pattern and the second coil pattern are electrically connected through a conductor in the substrate.
5. The communication device according to claim 1, wherein the coil pattern includes a first coil pattern and a second coil pattern, the first coil pattern has a first center point, the second coil pattern has a second center point, and the second center point is closer to the exposed surface than the first center point.
6. The communication device according to claim 1, further comprising a controller configured to display a message on the display to prompt an operator to adjust a position or orientation of the external device when a central axis of a coil in the external device is aligned with the substrate.
7. A communication device, comprising: a body; and wireless communication circuitry housed in

the body configured to communicate wirelessly with an external device proximate to the body through magnetic coupling, wherein the body is placed in an opening in a wall surface and fixed to the wall surface with an exposed surface of the body exposed from the wall surface, the wireless communication circuitry includes a substrate including a coil pattern to allow the magnetic coupling with the external device, the substrate is nonparallel to the exposed surface, the coil pattern includes a first coil pattern and a second coil pattern, and each of the first coil pattern and the second coil pattern includes at least one winding, the first coil pattern includes a first conductor located in a first direction and a second conductor located in a second direction, the first direction is one of an outward direction in which the exposed surface faces or a direction in which the body is placed, and the second direction is the other of the directions, the second coil pattern includes a third conductor located in the first direction and a fourth conductor located in the second direction, the first conductor and the third conductor are located on one surface of the substrate, and the second conductor and the fourth conductor are located on different layers in the substrate.
8. The communication device according to claim 7, wherein the substrate and the exposed surface form an angle of 60 degrees or more.
9. The communication device according to claim 7, wherein the substrate is perpendicular to the exposed surface.
10. A communication device, comprising: a body; and wireless communication circuitry housed in the body configured to communicate wirelessly with an external device proximate to the body through magnetic coupling, wherein the body is placed in an opening in a wall surface and fixed to the wall surface with an exposed surface of the body exposed from the wall surface, the wireless communication circuitry includes a substrate including a coil pattern to allow the magnetic coupling with the external device, the substrate is nonparallel to the exposed surface, the coil pattern includes a first conductor located in a first direction and a second conductor and a third conductor located in a second direction and connected parallel to the first conductor, the first direction is one of an outward direction in which the exposed surface faces or a direction in which the body is placed, and the second direction is the other of the directions, and the second conductor and the third conductor are located on different layers in the substrate.
11. The communication device according to claim 10, wherein the substrate is perpendicular to the exposed surface.